健康照護組

The association between ambient $PM_{2.5}$ exposure and suicide mortality rates in the general population and the mental illness population: A nationwide population-based study in Taiwan.

Zong-Rong Lee(李宗榮)[1]

[1] Department of Public Health, College of Health Sciences, Kaohsiung Medical University, Taiwan
Corresponding Information to E-mail: u109006205@gap.kmu.edu.tw Ext. 2315

Research Objective

- Suicide is a serious public health issue. In the SDGs, the suicide mortality rate is one of the indicators of Target 3.4: "By 2030, reduce premature mortality from non-communicable diseases by one third through prevention, treatment, and the promotion of mental health and well-being.
- Growing evidence indicates an association between air pollution and suicide. However, previous research findings have varied depending on the type of air pollutant, the study location, and the population, especially vulnerable populations in society.
- This study aims to examine an association between ambient $PM_{2.5}$ exposure and the suicide mortality rate in the general population and the mental illness population in Taiwan.

Materials and Methods

- **Data sources:** We conducted a retrospective ecological study at township level using the Taiwan National Health Insurance Research Database and the Taiwan Air Quality Monitoring Database from 2008 to 2019.
- **Outcome variable:** Suicide deaths were defined by ICD-10 codes, and information such as age, gender, and region of residence was obtained. The mental illness population was defined as those with a psychiatric diagnosis in the year prior to the suicide death, and we calculated age-adjusted death rates for suicide by gender across three periods (2008-2011, 2012-2015, and 2016-2019).
- **Exposure assessment:** A two-stage estimation procedure was used to estimate $PM_{2.5}$ concentrations using aerosol optical depth (AOD) estimates of surface $PM_{2.5}$, together with meteorological and land use variables. Specifically, a generalized additive model (GAM) was used in the first stage, and an XGBoost model was used in the second stage (Chen et al., 2023).
- **Statistical approaches:** We used the median $PM_{2.5}$ concentrations across three periods as thresholds to classify exposure into low and high levels, and used generalized estimating equations (GEE) to analyze the relationship between ambient $PM_{2.5}$ exposure and the suicide mortality rates in both populations.

Conclusions

- Our findings show that ambient $PM_{2.5}$ concentrations in Taiwan have a downward trend, and the age-adjusted death rate for suicide among adults in the general population and the mental illness population also has a downward trend.
- Across three periods, we find the evidence of the association between ambient $PM_{2.5}$ exposure and suicide mortality rate in the mental illness population, but not in the general population.
- It is necessary to focus on the influence of environmental exposure $PM_{2.5}$ on susceptible populations for further research.

Results

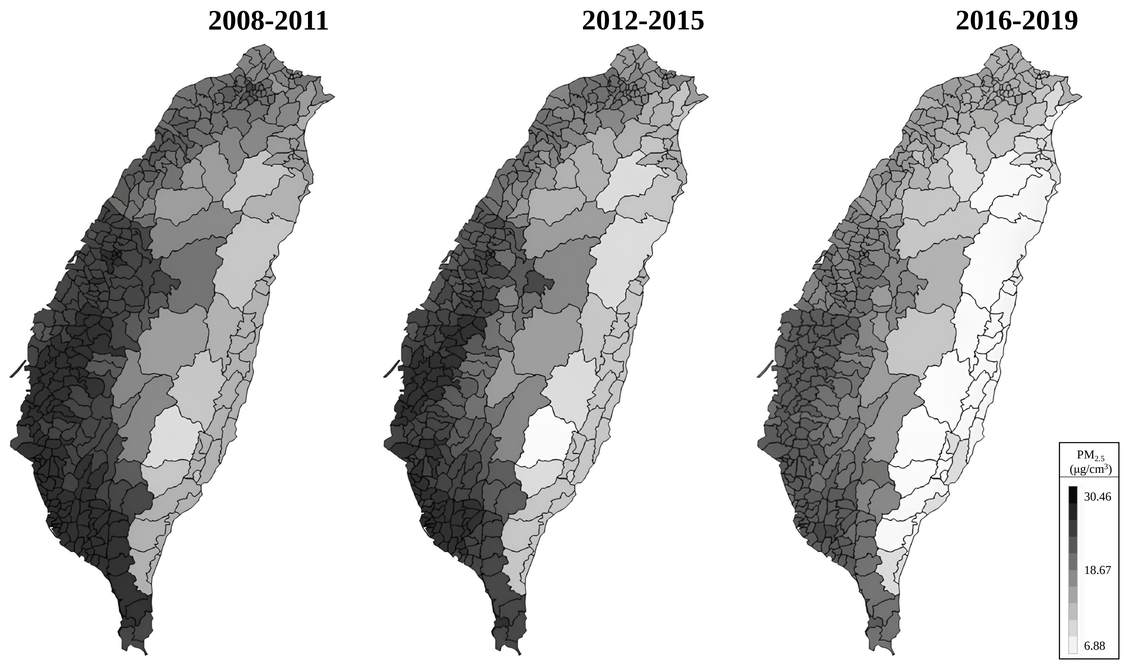


Figure 1. Distribution of ambient $PM_{2.5}$ concentrations in Taiwan.

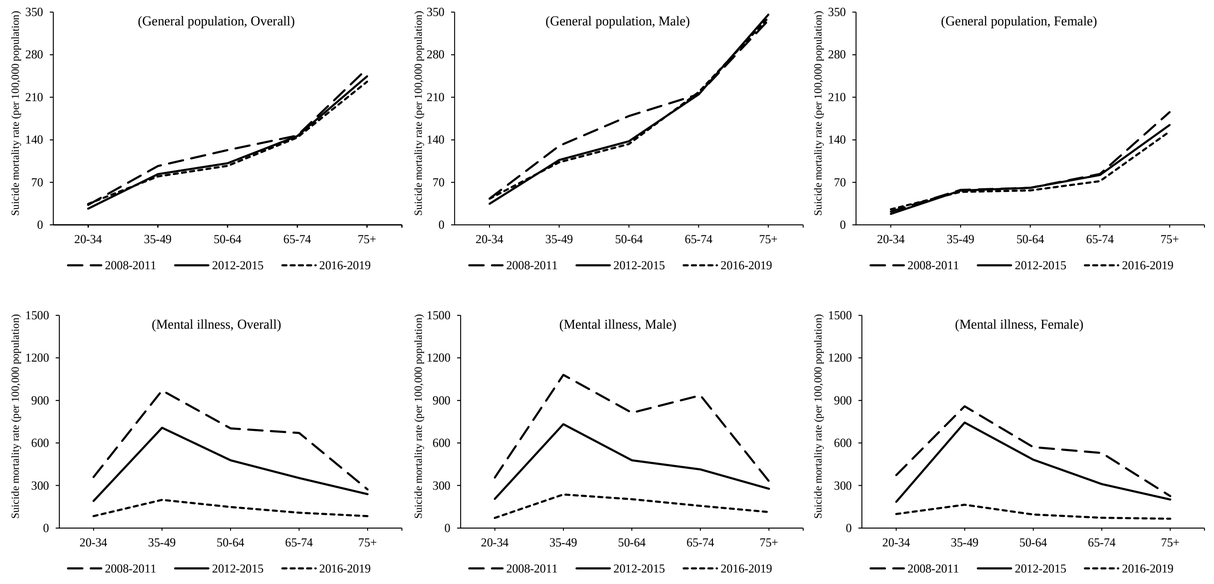


Figure 2. Trends in suicide mortality rates in both populations.

Table 1. Adjusted relative risk (aRR) of the suicide mortality rates using generalized estimating equations.

	Model 1		Model 2		Model 3	
	General aRR (95% CI)	Mental illness aRR (95% CI)	General aRR (95% CI)	Mental illness aRR (95% CI)	General aRR (95% CI)	Mental illness aRR (95% CI)
$PM_{2.5}$ (μg/m^3)	0.99 (0.99, 1.00)	**1.13 (1.10, 1.16)*** **	–	–	–	–
$PM_{2.5}$ level (ref: Low)						
High	–	–	0.97 (0.91, 1.03)	**1.90 (1.60, 2.25)*** **	–	–
Periods (ref: 2008-2011)						
2012-2015	–	–	–	–	**0.91 (0.87, 0.95)*** **	**0.67 (0.58, 0.77)*** **
2016-2019	–	–	–	–	**0.92 (0.88, 0.97)** **	**0.22 (0.19, 0.26)*** **
Sex (ref: Famle)						
Male	**2.21 (2.12, 2.31)*** **	**1.23 (1.07, 1.40)** **	**2.21 (2.11, 2.30)*** **	**1.27 (1.11, 1.44)*** **	**2.21 (2.11, 2.30)*** **	**1.32 (1.18, 1.47)*** **
Age group (ref: 20-34)						
35-49	**2.67 (2.47, 2.88)*** **	**3.17 (2.54, 3.95)*** **	**2.67 (2.47, 2.88)*** **	**2.99 (2.44, 3.67)*** **	**2.66 (2.46, 2.88)*** **	**2.79 (2.30, 3.37)*** **
50-64	**3.18 (2.95, 3.42)*** **	**2.17 (1.76, 2.68)*** **	**3.18 (2.95, 3.42)*** **	**2.07 (1.70, 2.52)*** **	**3.17 (2.94, 3.42)*** **	**1.93 (1.61, 2.32)*** **
65-74	**4.39 (4.05, 4.74)*** **	**1.77 (1.37, 2.29)*** **	**4.38 (4.05, 4.74)*** **	**1.80 (1.39, 2.34)*** **	**4.39 (4.05, 4.75)*** **	**1.62 (1.29, 2.02)*** **
75+	**8.01 (7.40, 8.67)*** **	1.01 (0.81, 1.26)	**8.01 (7.40, 8.67)*** **	0.95 (0.78, 1.15)	**8.00 (7.39, 8.67)*** **	0.96 (0.80, 1.16)
Region (ref: Northern)						
Central	0.99 (0.94, 1.06)	**0.56 (0.45, 0.69)*** **	0.98 (0.92, 1.04)	**0.78 (0.65, 0.93)** **	0.96 (0.92, 1.01)	1.07 (0.94, 1.21)
Southern	1.04 (0.96, 1.12)	**0.36 (0.28, 0.45)*** **	1.01 (0.94, 1.09)	**0.61 (0.51, 0.74)*** **	0.99 (0.94, 1.04)	1.03 (0.91, 1.17)
Eastern	**1.19 (1.04, 1.36)***	**2.82 (1.65, 4.83)*** **	**1.23 (1.08, 1.39)** **	1.55 (0.94, 2.56)	**1.23 (1.08, 1.40)** **	1.24 (0.84, 1.84)

*Significance at the 0.05 probability level. **Significance at the 0.01 probability level. ***Significance at the 0.001 probability level.
Model 1: adjusted for $PM_{2.5}$, sex, age group, region. Model 2: adjusted for $PM_{2.5}$ level, sex, age group, region. Model 3: adjusted for periods sex, age group, region.

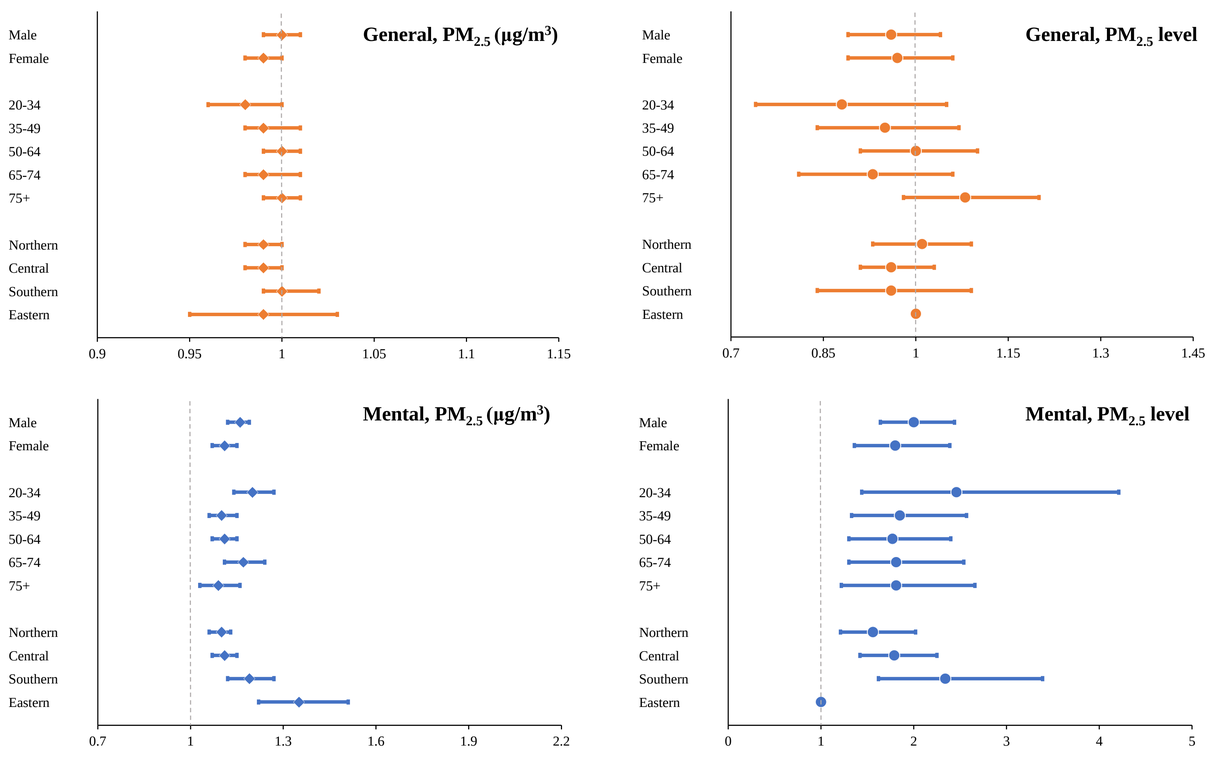


Figure 3. The forest plots of the aRR for exposure to $PM_{2.5}$ on the suicide mortality rate.